DEEP LEARNING · FINAL PROJECT REPORT

LAND COVER FROM SENTINEL-1 RADAR

A Deep-Learning Model, a Monthly-Change Prototype and an Honest Evaluation

Sentinel-1 VV and VH backscatter · FPN with an ImageNet-initialised ResNet-50
six land-cover classes at 10 m · Streamlit prototype with monthly change maps

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Data: Copernicus Sentinel-1 via Google Earth Engine · ESA WorldCover 2021 · CORINE Land Cover 2018

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Links and Resources

Item	What it contains	Public link
Dataset	Sentinel-1 monthly composites, labels and place lists of 372 places (about 80 GB; download single places, not the whole folder)	Google Drive folder
Model weights	Final model best_v4b_fpn_resnet50_imagenet_ctx.pt (about 100 MB) and stats_v4.json, folder model_weights/ of the dataset link	Google Drive folder
Earlier attempt (version 3)	Data, labels and weights of the first project stage	Google Drive folder (version 3)
Training code	Notebooks Deep_Learning_final_submission.ipynb (final) and Earlier_Attempt/ (first stage)	GitHub repository
Prototype code	prototype_v4/ (app.py, lulc_core_v4.py, visuals.py, visuals_data.py, README, requirements)	GitHub: prototype_v4
Demo video	Submitted separately on Moodle (full run for an urban area)	Moodle: Demo Video

Abstract

This project builds a land-cover classifier that uses **Sentinel-1 radar only** (VV and VH backscatter) and a prototype that turns a latitude, longitude and start month into twelve monthly land-cover maps with class shares and change. It was built in two stages. A **first attempt** (version 3) trained a model on 500 locations, but a diagnosis found four flaws in its data: an unintended incidence-angle offset in the radar values, images resized to pixels of about 39 m, 2025 images paired with 2021 labels, and almost no Urban pixels (1.1%). The **final project** corrected all four, sampled 370 places by class quotas, split them by geography, and added 72 training places for the weak classes after the first errors were seen.

The final model is an FPN with an ImageNet-initialised ResNet-50 encoder. It takes one month of VV and VH plus the annual mean and spread (six channels) and predicts six classes at 10 m. On 39 unseen test places it reaches an overall accuracy (OA) of **0.764** and a mean IoU (mIoU) of **0.633** (95% interval 0.545 to 0.702) against ESA WorldCover 2021; always predicting Forest scores 0.274. In two held-out regions it is weak: OA 0.587 in Australia, below a constant guess of 0.634, and 0.666 in South Asia. The prototype measures its own noise: on validation places a median of 2.9% of the pixels changes class within a year, so changes near 3% are within the noise. Every number in this report is measured, derived from a measured number, or labelled as a hypothesis.

1. Introduction

1.1 Task and research questions

Land-cover maps show what covers the ground: forest, grass, fields, buildings, bare soil or water. Optical satellites fail under clouds; radar satellites such as Sentinel-1 send their own microwave pulses and see through cloud and at night, but the echo (backscatter) is harder to interpret and carries speckle noise. The assignment asks for a deep-learning segmentation model that uses Sentinel-1 as its only input, and a prototype that takes a latitude, longitude and start month and returns twelve monthly maps, the class percentages per month, the month-to-month change and the total change in percentage points, and a 12-month visualisation. Four questions run through the report:

- Does adding the annual mean and spread of the radar signal ("context") help a one-month model?
- Does an ImageNet-pretrained encoder help although radar is not a photograph?
- How does the model behave in regions unlike the training data?
- How much of the change shown by a monthly map is real, and how much is classification noise?

1.2 Conventions

- Every number is **measured**, **derived** from a measured number, or labelled as a **hypothesis**.
- Intervals are 95% intervals from resampling whole places; differences between models are paired on the same places. The effective sample size is the number of places, not pixels.
- Decisions between models were made on validation places by rules written down before the results were seen. The word *earlier* in tables means an intermediate model of the final project; the *first attempt* is version 3.

Table 1: The project at a glance

Item	Value
Input	Sentinel-1 VV and VH, monthly median of up to five passes, 10 m, January 2020 to December 2021
Classes	Forest, Shrub/Grass, Cropland, Urban, Bare/Arid, Water/Wetland (from ESA WorldCover 2021)
Places	370: train 257, validation 41, test 39, held-out Australia 10 and South Asia 23
Final model	FPN, ImageNet-initialised ResNet-50, six input channels, 26.1 M parameters
Validation (41 places)	OA 0.772, mIoU 0.634 [0.563, 0.687]
Test (39 places)	OA 0.764, mIoU 0.633 [0.545, 0.702]
Held-out regions	Australia OA 0.587, South Asia OA 0.666
Noise floor of the prototype	median 2.9% of pixels change class per place-year (upper bound)
Independent check	Heidelberg 2020 against CORINE 2018: 78.8% agreement at 100 m

2. From the First Attempt to the Final Project

The project was built twice. The first attempt (version 3; notebook in the folder Earlier_Attempt, data and weights behind the second link above) gave a working model. Looking at its errors, especially for Urban, led to a diagnosis of the data and to a rebuild from the data upward. This chapter tells that path; Chapters 3 to 5 describe the final project.

2.1 The first attempt (version 3)

Version 3 drew 1,001 random locations stratified by biome region and selected 500 of them with a fixed class balance (125 Forest, 125 Shrub/Grass, 75 Cropland, 75 Bare/Arid, 50 Urban, 50 Water). For each location it downloaded the single Sentinel-1 passes of **2025** from Earth Engine and the ESA WorldCover **2021** labels, remapped to the same six classes. After dropping 27 locations with more than 80% unlabelled pixels, 448 locations remained. A UNet and an FPN were trained; adding the annual mean and spread of the radar signal as context raised the test mIoU of monthly composites from 0.429 to **0.526** (63 test locations, OA 0.712). But the Urban IoU stayed at 0.378 (validation 0.235), and in the held-out Australian region the context model reached an OA of only 0.486 against 0.813 for always predicting the biggest class.

2.2 What the diagnosis found

Table 2: Flaws of the first attempt and their correction

Flaw found	Effect	Correction in the final project
Unintended offset: the download function subtracted 2·cos(incidence angle) dB from VV and VH, although the Earth Engine GRD collection is already calibrated and terrain-corrected	Backscatter values shifted by an angle-dependent amount; the model learns distorted radar values	Removed; the collection is used exactly as delivered
The 10 km area was resized to 256 x 256 pixels, so one pixel covers about 39 m instead of 10 m	Small objects (buildings, roads, narrow rivers) disappear, which hurts Urban most	1,000 x 1,000 pixels at 10 m; the nine central 256 x 256 tiles are kept at native resolution
Sentinel-1 images of 2025 were paired with WorldCover labels of 2021	Four years of possible land change between image and label	Images of 2020 and 2021 with labels of 2021 (at most one year apart)
Urban was only 1.1% of the labelled pixels; some locations did not match their label (urban-tagged locations labelled water or bare, or mostly unlabelled)	Rare class under-trained; noisy training samples and thin evidence	Places chosen by WorldCover class shares with quotas (Urban 66, Peri-urban 24), at least 50% labelled pixels, class weights and a Dice loss

2.3 The final project and the 72 extra places

The final project was rebuilt with these corrections. Its first models (185 training places, FPN with context, trained from scratch) lifted the test mIoU to 0.617 and the **Urban IoU from 0.378 to 0.666**. Errors that remained on the validation places were Shrub/Grass (IoU 0.31, 27% of its pixels called Forest), Forest called Urban (10%), Cropland (0.55) and Water/Wetland (0.52), and in Australia the Forest IoU was only 0.008. Because the sampling was stratified from the start, the weak classes could be targeted: after these errors were seen, **72 further training places** were added by the same append-only rules (Shrub/Grass 16, Shrub-mixed 8, Peri-urban 12, Forest 12, Bare/Arid 8, Urban 8, Cropland 6, Water/Wetland 2). Validation, test and the two held-out regions did not change.

Table 3: Effect of the 72 extra places (same architecture and recipe, validation IoU per class)

Class	185 places	257 places	Change
Forest	0.650	0.688	+0.038
Shrub/Grass	0.306	0.375	+0.069
Cropland	0.549	0.596	+0.047
Urban	0.537	0.579	+0.042
Bare/Arid	0.729	0.701	-0.028
Water/Wetland	0.515	0.547	+0.032
mIoU	**0.548**	**0.581**	**+0.033 [-0.004, +0.069]**

Five classes improved and Bare/Arid lost a little; the mIoU gain is consistent but its interval includes zero. Each epoch also saw 39% more samples, so more data and more training steps are not separated. The last change was an ImageNet-initialised encoder (Chapter 5), which gave the final model.

Table 4: The three stages side by side (different test sets: compare the direction, not the digits)

Stage	Data	Pixel size	Test places	Model	Test mIoU	Urban IoU
First attempt (v3)	2025 images, 2021 labels	about 39 m	63	FPN + context	0.526	0.378
Final project, first models	2020-21 images, 2021 labels	10 m	39	FPN + context, 185 places	0.617	0.666
Final model	same, plus 72 places	10 m	39	FPN + context, ImageNet, 257 places	0.633	0.681

3. Data and Preprocessing

3.1 Overview

Table 5: Dataset at a glance

Property	Value
Radar source	Sentinel-1 GRD, interferometric wide mode, VV and VH, both orbit directions (Earth Engine, COPERNICUS/S1_GRD)
Regions	370 places spread over all continents; each a 10 km box (1,000 x 1,000 pixels at 10 m); the nine central 256 x 256 tiles (a 7.68 km square) are kept
Period	January 2020 to December 2021: 24 monthly composites per place
Images	27,695 for the first 300 places (median 108 per place) and 6,947 for the 72 extra places
Labels	ESA WorldCover v200 (2021), mapped from 11 to 6 classes
Usable places	370 (two of the first 300 had no Sentinel-1 image)

3.2 Sentinel-1 input and preprocessing

- **Calibration, dB conversion and terrain correction:** the Earth Engine collection is delivered thermal-noise-corrected, radiometrically calibrated, terrain-corrected and converted to decibels, as documented in the data catalogue. No further correction is applied; in particular the incidence-angle offset of the first attempt is gone.
- **Speckle:** reduced by combining several passes in time (per-pixel median of up to five passes per month); no spatial filter, so the 10 m detail is kept.
- **Image selection:** only images that cover the kept area; at most five passes per month, evenly spaced; an image needs at least 90% valid pixels, otherwise the nearest unused pass replaces it (at most nine tries).
- **No-data pixels** are stored as NaN, set to 0 after normalisation, and ignored by the loss and by every score. Each month is saved as a float16 array.

3.3 Labels and why six classes

The labels come from ESA WorldCover v200 (2021) and were remapped from its 11 classes to six when they were downloaded. Labels are used only for training and scoring, never at prediction time. The label map and the image share the same 10 m grid, so each pixel has a label without resampling.

Table 6: Mapping of the WorldCover classes to the six project classes

WorldCover code and class	Project class
10 Tree cover	Forest
20 Shrubland, 30 Grassland	Shrub/Grass
40 Cropland	Cropland
50 Built-up	Urban
60 Bare/sparse vegetation, 70 Snow and ice, 100 Moss and lichen	Bare/Arid
80 Water, 90 Herbaceous wetland, 95 Mangroves	Water/Wetland

Why the classes were merged. (1) Radar separates some of the 11 classes poorly: Shrubland and Grassland have a similar structure and similar backscatter. (2) Snow, moss, mangroves and herbaceous wetland are rare; in 370 places there are too few examples to train or evaluate them reliably, and a class with almost no test pixels makes the mIoU unstable. (3) The label products disagree most on exactly these classes, so finer classes would partly be learned from label noise. (4) The prototype reports class shares and changes, which need few, interpretable classes. The cost is that an mIoU over six merged classes is easier to reach than one over 11 classes and is not comparable with 11-class results.

Label quality. WorldCover reports a global accuracy of 76.7%. Dynamic World, an independent product, agrees with it on 76.5% of the pixels pooled over 12 places, but on only 28% of WorldCover's Shrub/Grass pixels; at Heidelberg, CORINE and WorldCover agree on 73%. Training only on pixels where two products agree was rejected: it would remove 72% of the Shrub/Grass pixels and hide the hardest class. The labels are model outputs, so label doubt caps what any model can score, most of all for Shrub/Grass.

3.4 Sampling, splits and class distribution

Sampling. Candidates are 10,000 random land points (looked up in the RESOLVE 2017 ecoregion map) plus 1,000 MODIS urban pixels (seed 2026). A place needs at least 20 Sentinel-1 passes in each of 2020 and 2021 and at least 50 in total, at least 50% labelled pixels, and a distance of at least 20 km to every other place. Places are stratified by the WorldCover class shares of their box, with quotas of Urban 66, Peri-urban 24, Shrub/Grass 48, Shrub-mixed 24, Cropland 30, Bare/Arid 24, Water/Wetland 18 and Forest 30, plus 36 class-blind random places that allow natural-frequency estimates. This gave 300 places, 298 usable; the 72 extra places of Section 2.3 bring the total to 370.

Splits. Places are split by 10-degree geographic blocks, so no block feeds two splits, and two regions are held out entirely: Australia (latitude -45 to -10, longitude 112 to 155) and South Asia (latitude 5 to 35, longitude 68 to 98). Heidelberg's block and a 100 km radius around it are excluded, so Heidelberg can serve as an independent case study. The training set has 257 places, validation 41, test 39, Australia 10 and South Asia 23.

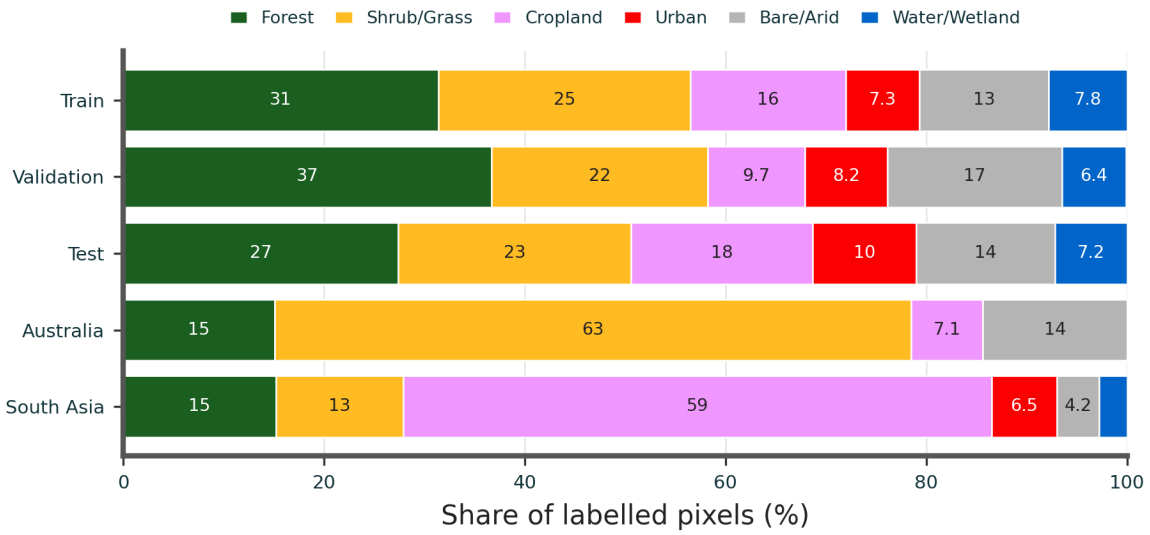


Figure 1: *Share of labelled pixels per class and split. The sampling is stratified, so these shares are not natural frequencies; Shrub/Grass is 63% of Australia and Cropland 59% of South Asia.*

3.5 Normalisation, patching, augmentation and class weights

- **Normalisation:** z-score with mean and standard deviation of the 185 original training places (VV -11.92 dB and 5.49; VH -19.29 dB and 5.80), kept frozen when the 72 places were added.
- **Context channels:** annual mean and standard deviation (divided by 3) of VV and VH over the calendar year of the month; with the month's own VV and VH they form six input channels.
- **Patching:** each of the nine 256 x 256 tiles of a place and each month with data is one patch, about 215 per place and 55,332 in the training pool.
- **Augmentation:** random horizontal and vertical flips and 90-degree rotations.
- **Class weights** (inverse square root of the class frequency, capped at 4): Forest 0.69, Shrub/Grass 0.76, Cropland 0.96, Urban 1.29, Bare/Arid 1.04, Water/Wetland 1.26.
- **Data volume:** raw download about 81 GB (65.5 GB measured for the first 300 places plus about 16 GB estimated), model-input cache 24.4 GB.

4. Model and Training

4.1 Design

- **Input, one month plus annual context.** A single month is ambiguous (a field, a meadow and a shrubland can look alike), so the model also receives how the place behaves over the year. On validation, context raised the mIoU from 0.502 to 0.548 (+0.046, interval includes zero).
- **Architecture, FPN with ResNet-50** (26.1 M parameters, segmentation_models_pytorch). The feature-pyramid decoder combines coarse and fine features, which suits class regions of very different sizes. UNet (32.5 M) and DeepLabV3+ (26.7 M) were tested with the same recipe and scored within 0.01 of the FPN, so the FPN, the fastest of the three (95 s per epoch against 113 s and 104 s), was kept.
- **ImageNet initialisation.** The encoder starts from torchvision's ResNet-50 weights; the first convolution expects three channels, so its filters are cycled over the six channels and scaled by 3/6. Because radar is not a photograph, this was treated as an open question and tested against a scratch twin with the identical recipe (Chapter 5).

4.2 Training setup

Table 7: Training recipe

Setting	Value
Loss	0.5 x class-weighted cross-entropy + 0.5 x Dice, ignoring pixels without data
Optimiser and schedule	AdamW, learning rate 3e-4, weight decay 1e-4, cosine schedule over at most 40 epochs
Batch and seed	batch size 16, seed 42
Samples per epoch	16 random (month, tile) patches per training place: 4,112 patches for 257 places
Model selection	epoch with the best mIoU on a fixed sample of 984 validation patches; early stopping after 10 epochs without improvement
Hardware	MacBook with Apple Silicon (PyTorch on MPS), 32 to 60 minutes per model

The saved final model is the epoch-7 checkpoint of a run that stopped at epoch 17. Replaying the seeded sampler shows that it had seen 22,506 of the 55,332 distinct training patches (40.7%), against 87.4% for the scratch twin at its best epoch: the pretrained model scored higher after seeing less than half as many patches. This fits the idea that pretraining speeds up early learning, which was not tested further; the counts measure exposure, not knowledge.

5. Experiments and Results

5.1 Protocol

- Every model is scored on all months with data and all tiles of the 41 validation places (579.8 million labelled pixels), without sampling pixels.
- Intervals resample whole places (2,000 draws); differences use the same resamples (paired).
- Model, epoch and pipeline were chosen on validation only, by rules written down before the numbers existed. The per-epoch scores printed during training use a fixed 984-patch sample, are optimistic, and were not used for decisions.

5.2 Comparison of model variants

Table 8: Validation results of all single models (41 places)

Model	Train places	Init	Epochs	OA	mIoU [95% interval]	Difference to FPN + context, 257 (paired)
FPN, monthly only (2 channels)	185	scratch	40	0.675	0.502 [0.424, 0.562]	-0.079 [-0.136, -0.030]
FPN + context	185	scratch	22	0.709	0.548 [0.460, 0.625]	-0.033 [-0.069, +0.004]
FPN + context, regularised	185	scratch	40	0.717	0.557 [0.464, 0.640]	-0.024 [-0.041, -0.003]
FPN + context	257	scratch	38	0.736	0.581 [0.489, 0.659]	reference
FPN + context, regularised	257	scratch	28	0.731	0.569 [0.484, 0.643]	-0.012 [-0.050, +0.033]
UNet-ResNet-50 + context	257	scratch	17	0.731	0.575 [0.493, 0.640]	-0.006 [-0.045, +0.033]
DeepLabV3+-ResNet-50 + context	257	scratch	26	0.732	0.572 [0.491, 0.642]	-0.009 [-0.042, +0.022]
FPN + context, ImageNet encoder (final)	**257**	**ImageNet**	**17**	**0.772**	**0.634 [0.563, 0.687]**	**+0.053 [-0.002, +0.101]**

"Regularised" means 30% context dropout, label smoothing 0.1, weight decay 1e-2 and an exponential moving average of the weights.

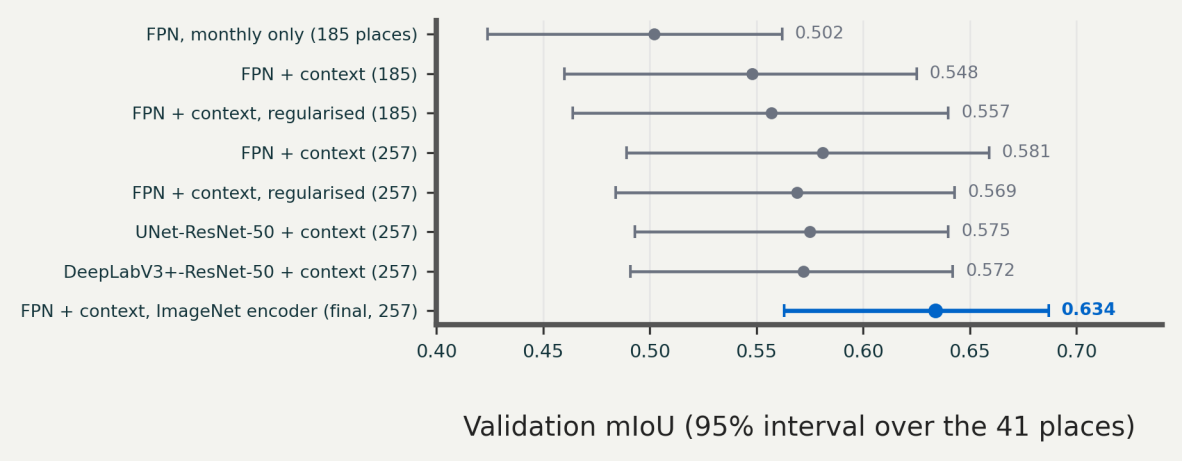


Figure 2: *Validation mIoU of all single models with 95% intervals over the 41 validation places.*

- **Pretraining** is the largest effect, +0.053 [-0.002, +0.101] over the scratch twin, and the only one that passed the bar set beforehand; its interval just touches zero. It helps Cropland (IoU 0.694 against 0.596), Water/Wetland (0.659 against 0.547) and Shrub/Grass (0.424 against 0.375).

- **Context channels:** +0.046 [-0.010, +0.094] over monthly-only; the first attempt gave the same direction (+0.096 [-0.006, +0.199] on its test set). Both intervals include zero, so the direction is consistent but not proven.
- **Extra places and architecture:** the 72 extra places gave +0.033 (Section 2.3); UNet, DeepLabV3+ and FPN differ by at most 0.009, well inside their intervals.
- **Ensembles, test-time augmentation, class offsets:** averaging the two scratch FPN models gave 0.576, no gain; adding the pretrained model to any set raised the best member by at most +0.009 (UNet + pretrained FPN, 0.643); four-flip test-time augmentation added +0.002; class offsets raised a two-model ensemble from 0.563 to 0.581 on validation but it scored 0.625 on test against 0.637 without offsets, so they were dropped.

Decision rules (written before the results). The ensemble of the two scratch FPN models would replace the earlier pipeline if it reached 0.581 on validation: it reached 0.576, no. The pretrained FPN as a single model would be adopted if it beat its scratch twin by 0.03 (0.611): it reached 0.634, yes. Any other ensemble had to beat a qualifying model by 0.01: the best was +0.009, no. A new architecture had to beat the FPN by 0.03: UNet and DeepLabV3+ did not. The final pipeline is therefore one model, the ImageNet-initialised FPN, with no ensemble, offsets or test-time augmentation.

5.3 Test-set results

Table 9: Test-set scores (39 places)

Model	OA	mIoU [95% interval]	Urban IoU	Final minus this model (paired)
Final: pretrained FPN + context	0.764	0.633 [0.545, 0.702]	0.681	not applicable
FPN + context, 185 places (earlier)	0.747	0.617 [0.529, 0.679]	0.666	+0.016 [-0.017, +0.058]
Two models averaged with offsets (earlier)	0.758	0.625 [0.539, 0.688]	0.691	+0.008 [-0.032, +0.051]
Majority-class guess (always Forest)	0.274	not applicable	not applicable	not applicable

The final model's test mIoU (0.633) matches its validation mIoU (0.634). The earlier models scored higher on test than on validation (0.617 against 0.548), so the gaps are smaller on the test set and the improvements over them are not significant.

Table 10: Per-class scores of the final model on the test set

Class	IoU	Precision	Recall	F1	Share of test pixels (%)
Forest	0.758	0.857	0.868	0.862	27.4
Shrub/Grass	0.481	0.588	0.725	0.650	23.2
Cropland	0.619	0.840	0.701	0.764	18.1
Urban	0.681	0.712	0.940	0.810	10.3
Bare/Arid	0.528	0.876	0.570	0.691	13.8
Water/Wetland	0.732	0.934	0.772	0.846	7.2

The mean F1 is about 0.77. Shrub/Grass and Bare/Arid are the weakest classes. Urban is found almost completely (recall 0.940) but with more false alarms (precision 0.712); Water/Wetland has the opposite profile. A natural-frequency check on the 27 test places from the random land frame gives OA 0.794 and mIoU 0.596 [0.473, 0.705]; with 27 places this is thin evidence.

5.4 Confusion matrices and held-out regions

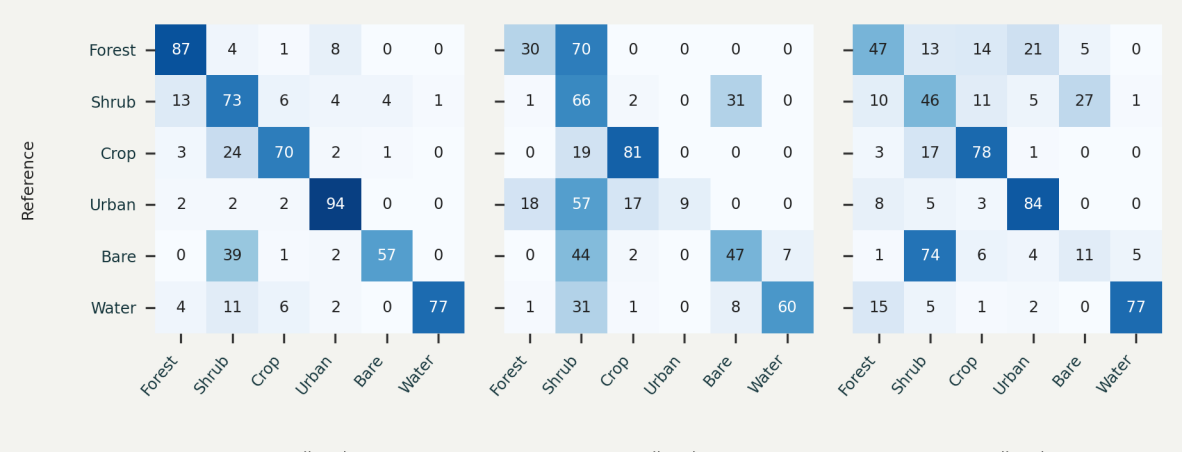


Figure 3: *Confusion matrices of the final model on the test set and on the two held-out regions (percent of each reference class; rows sum to 100).*

On the test set the largest errors are Bare/Arid called Shrub/Grass (39%), Cropland called Shrub/Grass (24%), Shrub/Grass called Forest (13%) and Forest called Urban (8%). The held-out regions have extreme class mixes, so overall accuracy is compared with the constant guess that predicts the biggest class.

Table 11: Held-out regions (OA / mIoU)

Region	Places	Final model	FPN + context, 185 places	Two models + offsets	Constant guess (OA)
Australia	10	0.587 / 0.291	0.501 / 0.228	0.470 / 0.205	0.634 (always Shrub/Grass)
South Asia	23	0.666 / 0.417	0.638 / 0.368	0.660 / 0.418	0.586 (always Cropland)

Australia: the final model is below the constant guess. Only 30% of Forest pixels are found (70% are called Shrub/Grass) and 31% of Shrub/Grass is called Bare/Arid; Cropland works (IoU 0.667). Urban and Water cover 0.04% and 0.02% of the pixels and cannot be interpreted. The gain over the 185-place model (+0.062 [-0.016, +0.129]) is not significant. **South Asia:** the gain over the 185-place model is clear (+0.049 [+0.012, +0.082]) and equal to the two-model pipeline; Cropland (IoU 0.728) and Water/Wetland (0.661) work, while Bare/Arid is almost lost (IoU 0.054, 74% called Shrub/Grass).

5.5 Training curves

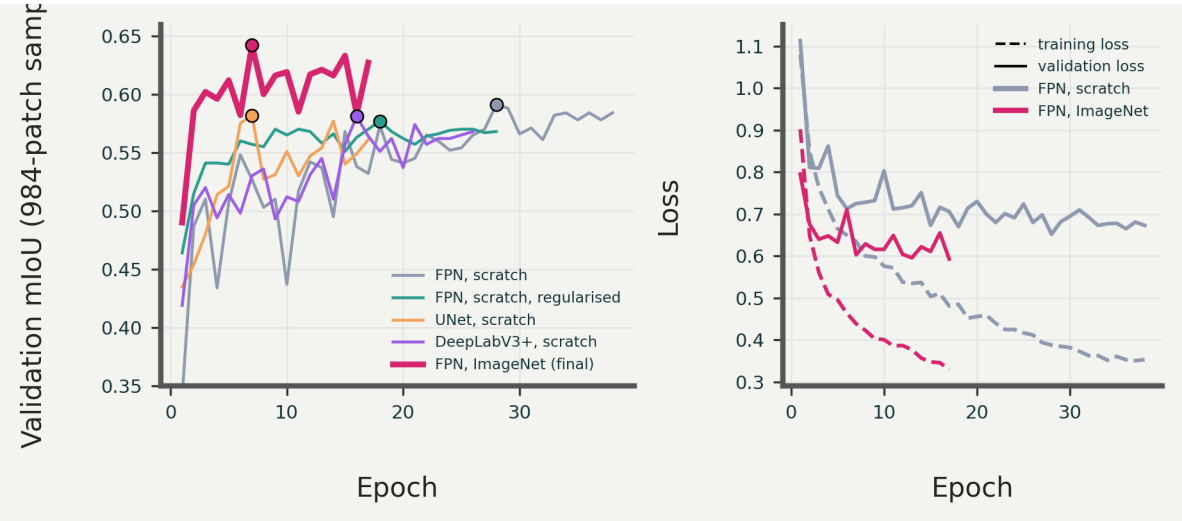


Figure 4: *Left: validation mIoU per epoch of the five models of Section 5.2 (dot = saved epoch). Right: training and validation loss of the scratch FPN and the pretrained FPN. Values are on the fixed 984-patch validation sample.*

The scratch models climb slowly and noisily; their best epochs are 28, 18, 7 and 16 (FPN, regularised FPN, UNet, DeepLabV3+). The pretrained model is the highest curve at every one of its 17 epochs, reaches 0.59 at epoch 2 and then stays between 0.58 and 0.64. Its best epoch is 7; afterwards the training loss keeps falling (0.90 to 0.33) while the validation loss stays flat between about 0.59 and 0.71, a widening gap that signals overfitting, and early stopping ended the run at epoch 17. These values use the 984-patch sample, are optimistic, and differ from the full-validation scores of Section 5.2.

6. Prototype

The Streamlit prototype rebuilds the training-data recipe for any place and start month, predicts the final model on the whole 7.68 km area, and reports shares, changes and how much of the change is likely noise. It ran end to end on real Earth Engine data for Heidelberg (2020 and 2025), Auckland (2025) and Singapore (April 2025 to March 2026); the rest was tested offline on synthetic data. Source code and README are behind the prototype link above.

6.1 System architecture, data retrieval and monthly compositing

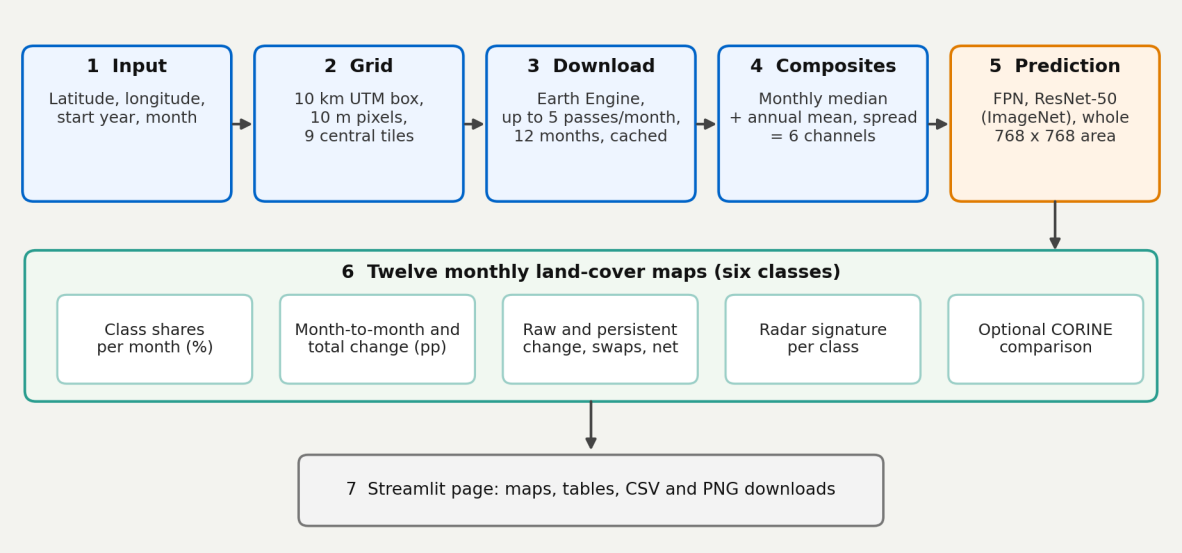


Figure 5: *Pipeline of the prototype from the typed place to the twelve maps and the derived analyses.*

- **Input:** latitude, longitude, start year and month; the app checks that all 12 months are in the past and warns outside 2020 and 2021, the years the model was trained on.
- **Grid and download:** a 10 km box on an exact UTM grid at 10 m with the nine central tiles; for each of the 12 months at most five evenly spaced passes, with the same selection rules as in training. Months are cached on disk, so a second run takes seconds.
- **Composites and context:** the per-pixel median per month, and the mean and standard deviation of the 12 months as context. For a January start this equals the training recipe; for other start months it is an approximation that ran once without error (Section 7.2) and was not compared with a January start.
- **Prediction and outputs:** the final model on the stitched 768 x 768 area in one pass; twelve monthly maps, class shares, month-to-month and total change, persistent change, a radar signature per class, an optional CORINE comparison, and CSV and PNG downloads.

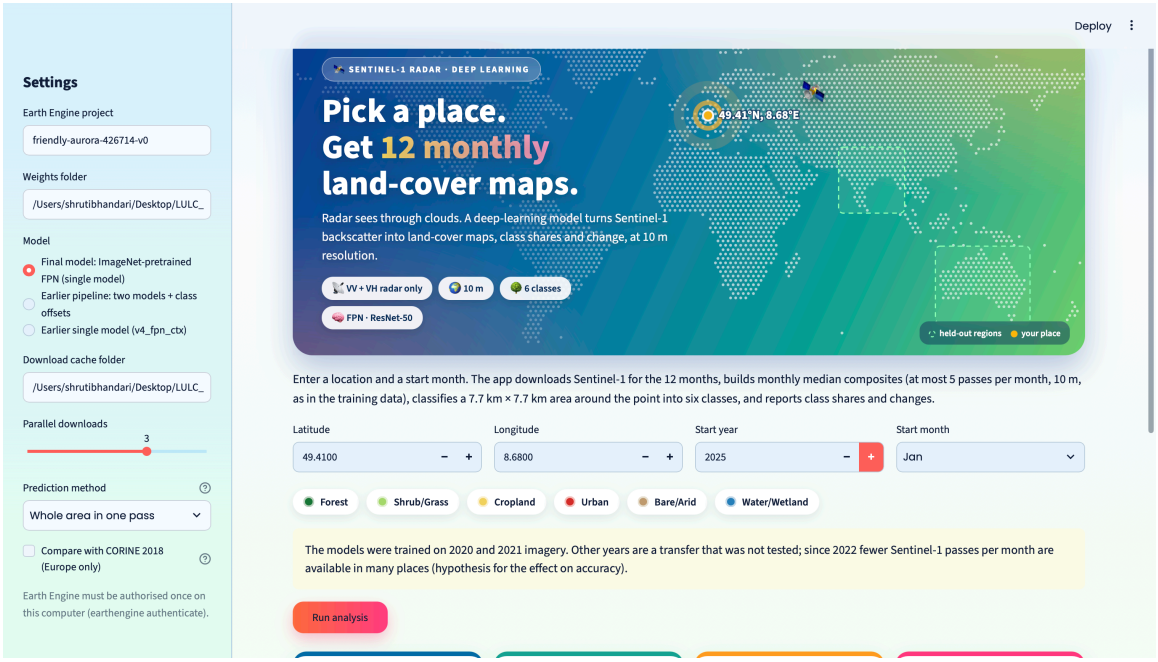


Figure 6: *The prototype start page: banner with a world map whose pin follows the typed coordinates, the input fields, the class legend and the notice about untested years.*

6.2 How class percentages and changes are calculated

- **Share:** pixels of a class divided by the pixels with data in that month, times 100. A month without a usable image is skipped, not counted as zero.
- **Month-to-month change:** the share of a month minus the share of the previous month, in percentage points. **Total change:** last month with data minus first.
- **Raw pixel change:** the percentage of pixels whose class differs between the first and the last month. It counts classification errors and seasonal effects as change, so it is an upper bound.
- **Persistent change:** the most frequent class of the first three months with data is compared with that of the last three (adjustable from one to four); a pixel counts as changed only if both are stable and different.
- **Two-way swaps and net change:** pixels that move from class A to B are matched against pixels that move from B to A. The matched pixels are swaps, typical of jitter along class boundaries; the rest is the net one-way conversion, in pixels and hectares (one pixel is 0.01 ha).

6.3 Radar signature

For each class the app plots the mean VV and VH backscatter of the pixels the model assigns to it, month by month, with a band of one standard deviation. At Heidelberg, VV separates Urban (about -6 to -7 dB) from Forest (about -9 dB) while VH barely separates them; Shrub/Grass, Cropland and Forest overlap strongly, and Cropland shows a seasonal dip in April and May. The chart describes what the model calls each class; it is not independent ground truth, because the same radar values drove the prediction.

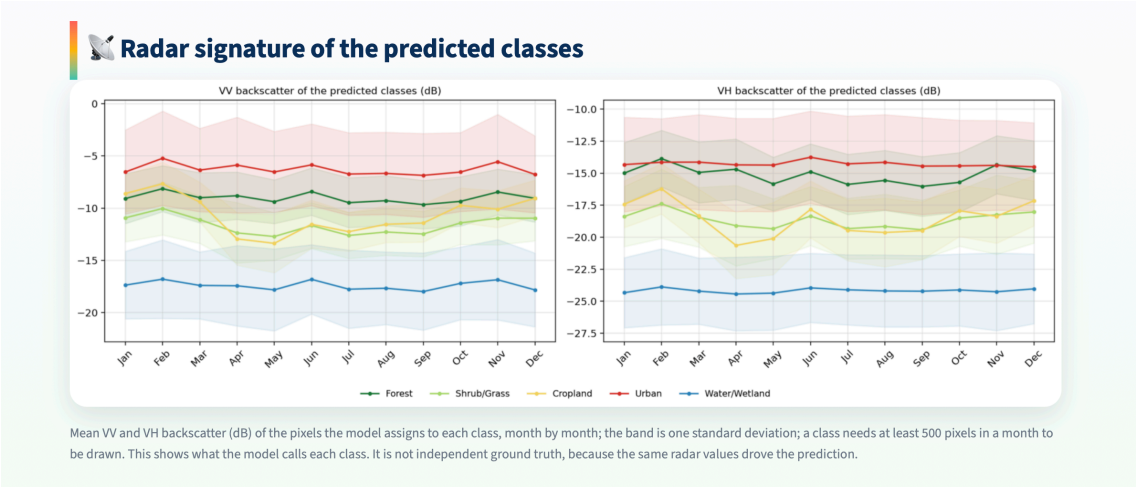


Figure 7: *Radar signature of the predicted classes at Heidelberg 2020 (mean VV and VH per month, band of one standard deviation).*

6.4 Tile seams and how much of the change is noise

The models were trained on separate 256 x 256 tiles, so predicting each tile alone leaves seams along the borders of the 3 x 3 grid: on the validation places the predicted class changes across an interior tile border for 18.7% of adjacent pixel pairs, against 1.3% elsewhere (seam excess 17.4 percentage points). Predicting the whole area in one pass cuts the excess to 0.14 points with an mIoU change of +0.0045 [-0.0073, +0.0184]; overlapping blended windows give 0.16 and +0.0119. A rule fixed before the test (at most 0.005 mIoU lost and at least 50% less seam excess; the lower excess wins) selected the whole-area method as the default, and overlapping windows remain selectable. One caveat: whole area changes about 7% of the pixels even far from any border, probably because the network never saw 768 x 768 inputs. All test and region scores in this report were computed tile by tile.

To measure false change, the final model was run on the 41 validation places for 2020 and 2021 (82 place-years). These places contain real change too, so the values are upper bounds for false change.

Table 12: Change measures over the 82 place-years (median / 90th percentile / maximum)

Measure	Tile by tile	Whole area (default)	Overlapping windows
Largest monthly move of any class (pp)	2.93 / 13.92 / 52.43	2.55 / 10.86 / 65.07	2.63 / 12.59 / 55.17
Raw first-to-last change (% of pixels)	3.99 / 10.65 / 27.89	2.86 / 8.53 / 36.56	3.95 / 9.56 / 28.16
Persistent change (% of stable pixels)	3.14 / 10.53 / 18.12	2.58 / 7.26 / 15.91	3.02 / 9.88 / 18.02

A typical place-year shows about 3% of the pixels changing class, so a result near 3% is within the noise; in 90% of place-years the raw change stays below 8.5%. The persistent vote lowers the median by only 10% to 24%: it refines the number but does not remove the noise. The worst place-year moved a class by 65 percentage points in one month; whether such cases are real events (flooding, snow, burnt land) or artifacts was not separated. The prediction method adds no false change (paired median difference of the persistent change -0.071 [-0.355, +0.002] pp for whole area against tile by tile).

Not tested: a place with a known real change (flood, harvest, construction), so a quiet result does not prove that real change would show up; years other than 2020, 2021 and the three 2025 runs; start months other than January beyond one April run; a fresh installation and Earth Engine accounts other than the author's.

7. Case Study and Further Runs

7.1 Heidelberg 2020 against CORINE 2018

Set-up. 49.41 N, 8.68 E, January to December 2020, final model, whole-area prediction, 60 Sentinel-1 images (five per month). Heidelberg lies in the zone excluded from the training pool. The reference is CORINE Land Cover 2018, a 100 m map with a 25 ha minimum mapping unit that was not used for training. The 44 CORINE classes were mapped to the six project classes by a fixed rule (artificial surfaces to Urban; forests to Forest; arable land, permanent crops and mosaics to Cropland; pastures, natural grassland, heath, scrub and green urban areas to Shrub/Grass; mines, beaches, rock, sparse vegetation and burnt areas to Bare/Arid; wetlands and water to Water/Wetland); mixed classes are a judgement call. The most frequent class over the 12 months per pixel is compared at 100 m and at 10 m.

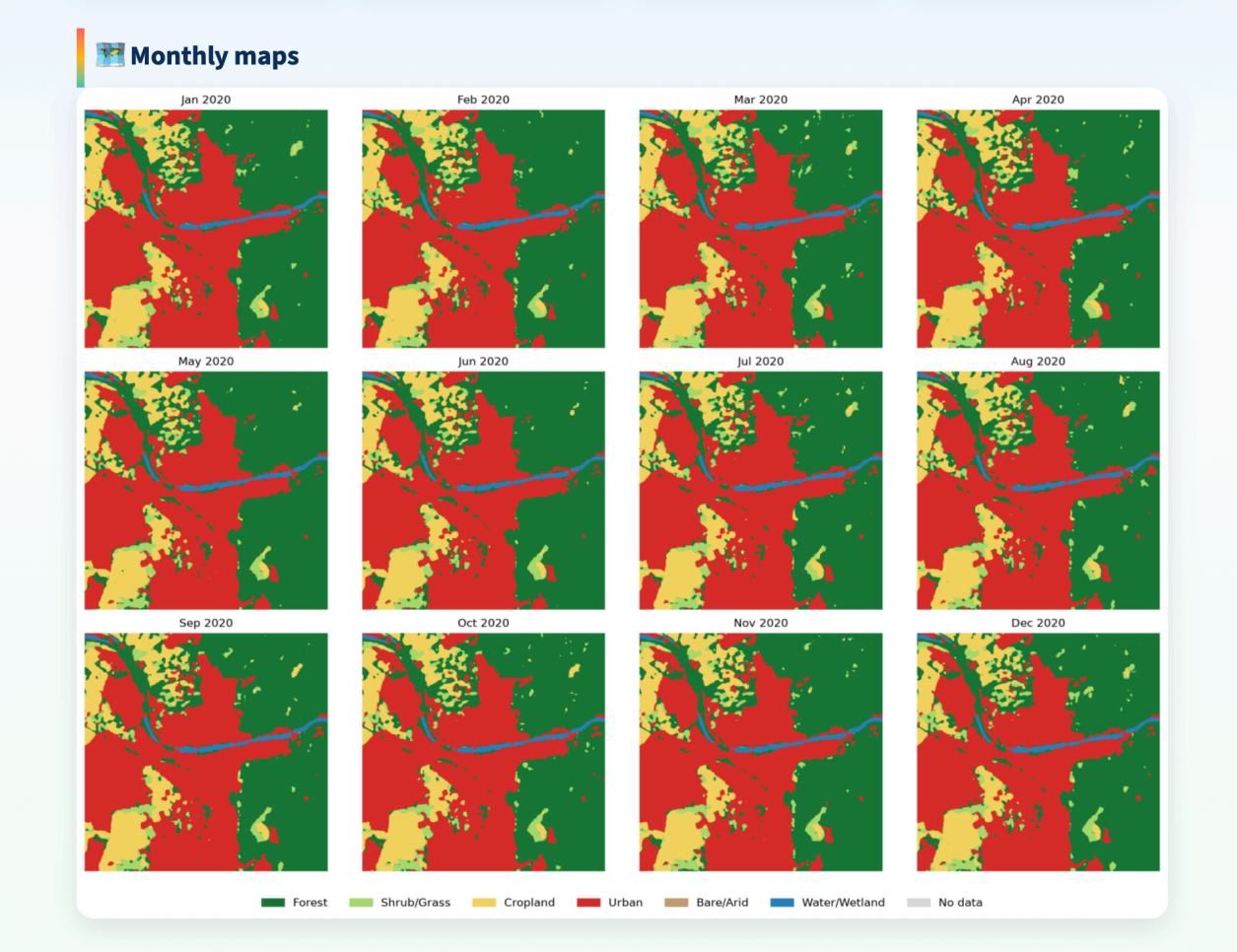


Figure 8: *The twelve monthly maps of Heidelberg 2020 produced by the final model.*

Agreement. 78.8% at 100 m (mIoU 0.488) and 78.6% at 10 m; tile-by-tile prediction gave 79.9% and 0.490.

Table 13: Heidelberg 2020: shares and agreement per class (CORINE at 100 m)

Class	CORINE share (%)	App share (%)	IoU	Precision	Recall
Forest	30.8	42.2	0.710	0.719	0.983
Shrub/Grass	4.3	3.7	0.067	0.136	0.116
Cropland	21.9	11.1	0.481	0.966	0.489

Class	CORINE share (%)	App share (%)	IoU	Precision	Recall
Urban	40.1	41.6	0.791	0.868	0.900
Bare/Arid	0.0	0.0	not defined	not defined	not defined
Water/Wetland	2.9	1.4	0.393	0.843	0.424

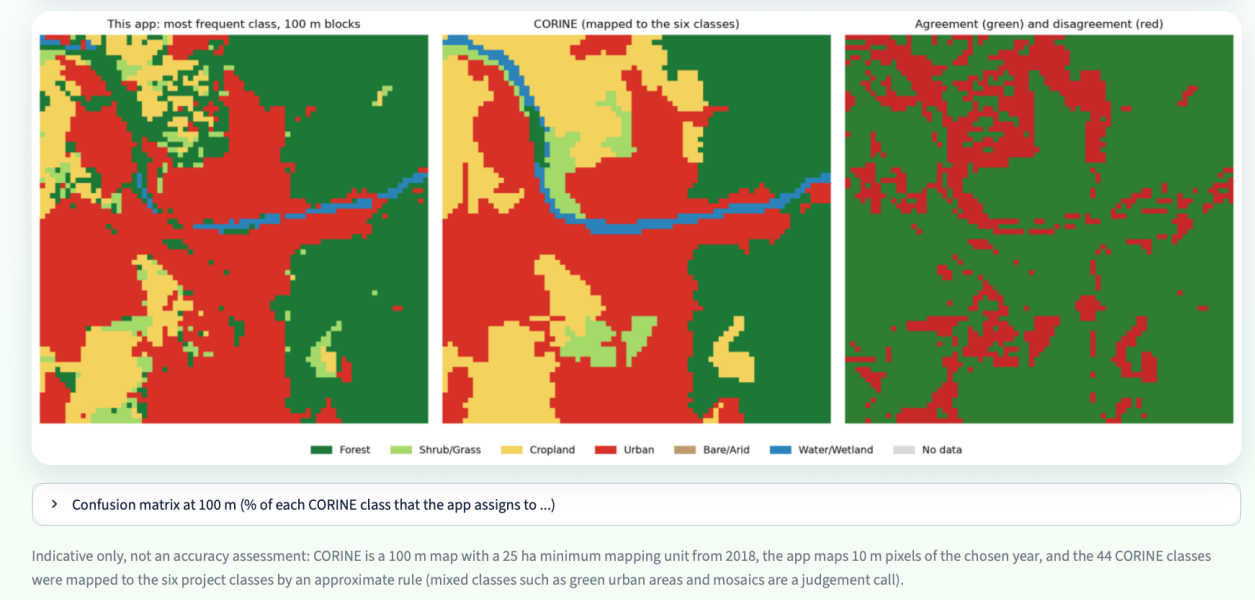


Figure 9: *The app map, the CORINE map mapped to six classes, and the agreement map at 100 m (green agrees, red disagrees).*

- Urban agrees well (IoU 0.791) and the app's Urban share (41.6%) is close to CORINE's (40.1%).
- Forest is found almost completely (recall 0.983) but the app's Forest share is 11 points above CORINE's (precision 0.719).
- Cropland is almost always right when predicted (precision 0.966) but half of CORINE's Cropland is missed (recall 0.489); a likely reason (hypothesis) is that CORINE's mixed cultivation classes, which the mapping counts as Cropland, look like Forest or Urban to radar.
- Water/Wetland recall is 0.424: the Neckar is a few pixels wide and CORINE draws rivers at least 100 m wide. Shrub/Grass almost never overlaps (IoU 0.067), in line with the label disagreement on this class (Section 3.3).

The check has four limits: CORINE is two years older than the imagery, it has a coarser resolution and minimum mapping unit, the class mapping is approximate, and it is one place. It is an indicative check, not an accuracy assessment.

Change over 2020. Class shares stay close to constant (January 42.0% Forest, 4.7% Shrub/Grass, 10.7% Cropland, 41.1% Urban, 1.5% Water/Wetland; December 41.9, 5.3, 10.3, 41.0 and 1.5): the largest monthly move is 2.0 points and no class changes by more than 0.6 points between January and December. 4.0% of the pixels change class (raw), 2.9% of the stable pixels change persistently, and 80% of those are two-way swaps; the net one-way change is 0.58%. The largest net flow, Forest to Urban (4,299 pixels one way against 2,615 the other, net 16.8 ha), is not validated: with tile-by-tile prediction the pair was almost balanced (2,745 against 2,659), so part of the asymmetry may come from the prediction method or from seasonal differences (hypotheses), not from urbanisation.

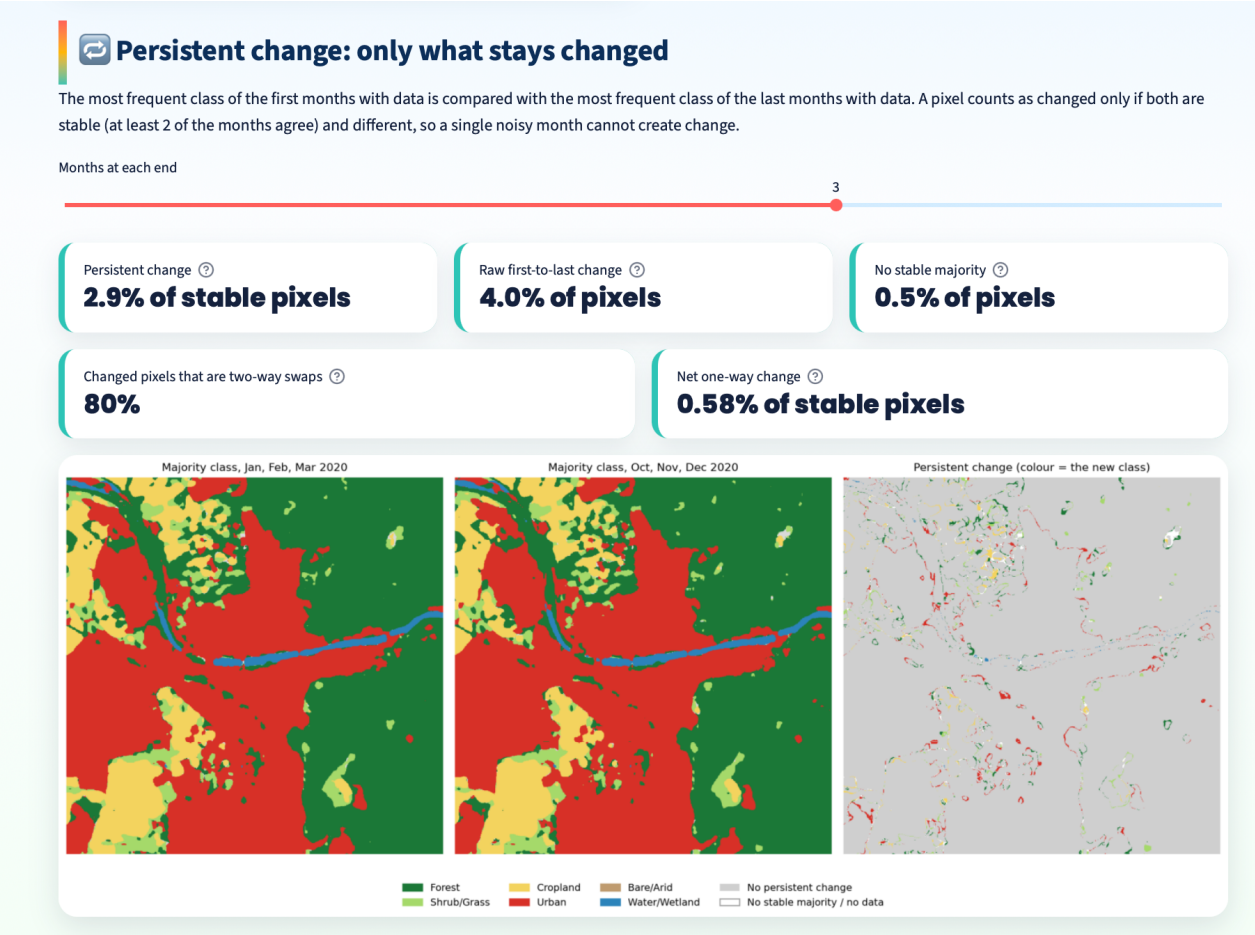


Figure 10: *Heidelberg 2020: persistent change with three months at each end (first three months, last three months, and the pixels that stay changed).*

7.2 Further runs: Heidelberg 2025, Auckland 2025, Singapore

Table 14: The four runs of the prototype

Run	Images	Main shares (%)	Largest monthly move (pp)	Raw change (%)	Persistent (%)	Two-way swaps (%)	Net one-way (%)
Heidelberg 2020	60	Forest 42, Urban 41, Cropland 11	2.0	4.0	2.9	80	0.58
Heidelberg 2025	60	Forest 43, Urban 40, Cropland 10	4.4	3.9	3.2	63	1.18
Auckland 2025	60	Urban 65, Water 30	1.1	2.2	1.6	70	0.50
Singapore Apr 2025 to Mar 2026	49	Urban 50, Forest 42, Water 6.5	3.4	3.8	1.5	88	0.18

- **Heidelberg 2025** (does the model transfer to a later year?): maps and shares are plausible and no class differs by more than 1.8 points from January 2020. CORINE agreement fell from 78.8% to 77.1% (mIoU 0.474), which is not evidence of drift because a seven-year gap lowers agreement even for a perfect map. Change is noisier; the largest net flows are Forest to Shrub/Grass (24.8 ha) and Forest to Urban (20.0 ha). Whether they are real change, seasonal effects or drift cannot be separated with one place and one year; they are hypotheses.
- **Auckland 2025** (southern hemisphere, very different class mix): no reference map is available (CORINE covers Europe only), so this is a plausibility and robustness check. Water/Wetland is about

30%, Urban about 65%; change is small. The VV backscatter of Water/Wetland rises by about 3 dB from November to December (hypothesis: wind roughening of the water).

- **Singapore, April start** (tests a non-January start): only 49 images because several months have two or three (April to March: 4, 4, 2, 3, 4, 5, 5, 5, 5, 4, 5, 3). The three months with only 2 or 3 images (June, July, March) show Forest at 39.6 to 39.7% and Urban at 52.8 to 53.0%, against 42.2% and 50.2% on average for the nine months with 4 or 5 images: about 2.6 points less Forest and 2.7 points more Urban. A likely explanation (hypothesis) is that a median of two or three images keeps more speckle, which the model reads as built-up surface. The raw first-to-last change of 3.8% (Forest -3.1, Urban +3.2) therefore mostly mimics change; the persistent vote gives 1.5% and a net one-way change of only 0.18%. The app now shows a notice for months with fewer than four images.

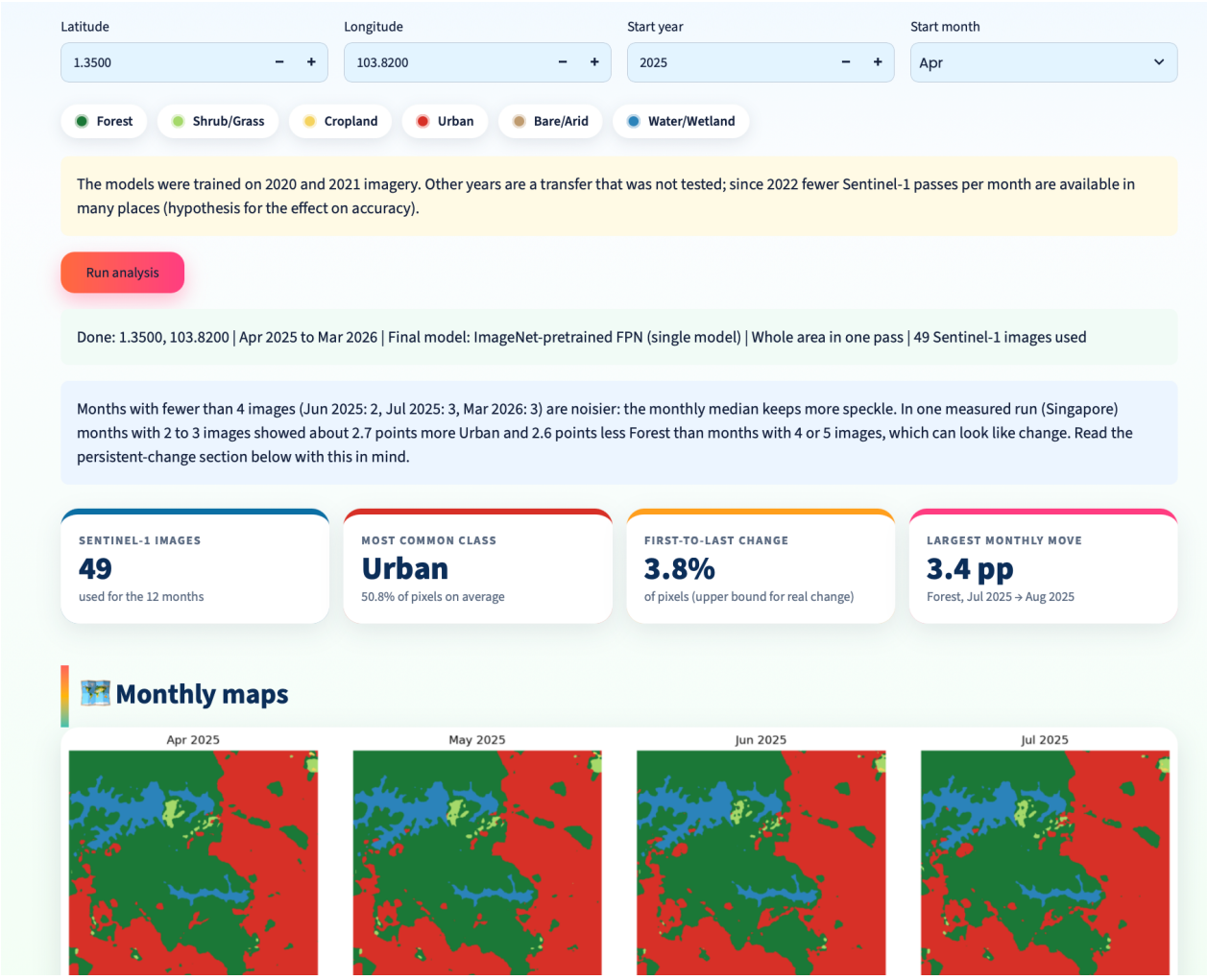


Figure 11: *Singapore: the notice for months with fewer than four images, the four summary cards and the first monthly maps.*

8. Critical Discussion

The results are solid for places similar to the training data and weak elsewhere, and the evidence has limits that the scores alone do not show. Each limit is marked as measured or as a hypothesis.

8.1 Regions unlike the training data

Measured: in Australia the final model reaches an OA of 0.587, below the 0.634 of always predicting Shrub/Grass; it finds only 30% of Forest (70% is called Shrub/Grass) and sends 31% of Shrub/Grass to Bare/Arid. In South Asia Bare/Arid is almost lost (IoU 0.054, 74% called Shrub/Grass). **Hypothesis:** arid shrubland and bare ground, and sparse forest and shrub, are close in radar backscatter, and the training places hold few examples of these landscapes. The earlier models fail in the same way (OA 0.501 and 0.470 in Australia), which points to the data and not to one model; no experiment tested the causes. An Australian-type place should not be trusted for Forest or Shrub/Grass shares.

8.2 Confusion between classes in radar

Measured (test set): Bare/Arid called Shrub/Grass 39%, Cropland called Shrub/Grass 24%, Shrub/Grass called Forest 13%, Forest called Urban 8%. In the radar signature at Heidelberg, the VH backscatter of Urban and Forest is almost the same (about -14.5 to -16 dB) and VV carries their separation; Shrub/Grass, Cropland and Forest overlap strongly within one standard deviation. Because these are the model's own classes, this shows what the model uses, not independent evidence. **Hypothesis:** with VV and VH alone, vegetation classes of similar structure cannot be separated by one month of data, which is why the annual context helps.

8.3 Seasonal effects and the number of passes that look like change

A monthly map classifies from backscatter, which also depends on crop cycle, soil moisture, snow, wet leaves and the number of passes in the month. The measured noise floor is a median of 2.9% of pixels changing class per place-year (whole area), with a 90th percentile of 8.5% and a worst case of 36.6%; the median largest monthly move is 2.6 percentage points, with a worst case of 65. These places contain real change as well, so the numbers are upper bounds, and the causes of the worst cases were not separated. A different number of passes per month can also mimic change: in Singapore the months with 2 or 3 images showed about 2.7 points more Urban and 2.6 points less Forest (Section 7.2), observed in one run only.

8.4 How classification errors inflate change percentages

Every pixel that is wrong in one month and right in the next counts as change. In a simulation with an independent error of 15% per month, a pixel that never changes shows a raw first-to-last change of about 25%, and the persistent vote still leaves about 11%. In the real data the vote lowers the median by only 10% to 24%, because errors persist for several months and 80% of the persistently changed pixels at Heidelberg are two-way swaps along class boundaries. The net columns are the more careful estimate of one-way conversion.

8.5 Limits of the evidence

- **Sample size:** 39 test places give intervals of about +/-0.07 mIoU; most differences between models have intervals that include zero, so only the direction is reported.

- **Test set and selection:** the test set was looked at three times (first single model, ensemble with offsets, final model). The final model was chosen on validation by rules fixed beforehand, a weaker guarantee than one single look. The validation places also chose the epoch, the architecture and the pipeline, so validation scores are somewhat optimistic.
- **Single seed:** each configuration was trained once, so run-to-run variation is not measured.
- **Six merged classes and tile-by-tile scores:** the mIoU is not comparable with an 11-class mIoU, and what the merge hides inside Shrub/Grass, Bare/Arid and Water/Wetland was not analysed. All scores were computed tile by tile and include the seam effect that the prototype removes.
- **Labels and years:** the labels are model outputs; training used 2020 and 2021 and only three 2025 runs were checked.
- **Stages are not comparable:** the first attempt and the final project use different places, test sets and class balances, so their scores show a direction only.

9. Conclusion

A radar-only model with annual context and an ImageNet-initialised encoder classifies six land-cover classes at 10 m with an OA of 0.764 and an mIoU of 0.633 [0.545, 0.702] on 39 unseen places, far above a majority-class guess (0.274). The path to it was a diagnosis, not a tuning exercise: the first attempt had an unintended offset, 39 m pixels, mismatched image and label years and too little Urban; after the rebuild, the Urban IoU rose from 0.378 to 0.681. The four corrections were made together, so their individual effects were not measured. Pretraining is the one model change that clearly lifts the validation score (+0.053), the architecture does not matter, and the model does not generalise to Australia. The prototype produces twelve monthly maps and change numbers for any place and start month; changes near 3% are within its measured noise.

Table 15: Research questions and answers

Question	Answer from the measurements
Does the annual context help?	Probably: +0.046 mIoU on validation (+0.096 in the first attempt); both intervals include zero
Does ImageNet pretraining help radar input?	Yes, the largest effect: +0.053 mIoU, interval just touches zero; the reason is a hypothesis
Do architecture, ensembles or the 72 extra places matter?	Architecture and ensembles: no (within 0.01); extra places: +0.033, not significant
How does the model behave in unseen regions?	Poorly: Australia is below a constant guess; South Asia loses Bare/Arid
How large is the false change?	Median 2.9% of pixels per place-year (upper bound); a different number of passes per month can mimic change

Lessons learned and future work

- **Diagnose the data before tuning the model:** the offset, the 39 m pixels and the year mismatch cost more than any architecture choice could recover.
- **Fix the decision rules before the numbers exist:** it made every later choice easy to defend and showed where the evidence is thin.
- **Protect destructive steps:** an accidental "Run All" on the submission day deleted the dataset folder on Google Drive. It was restored from the Drive bin, and the three notebook cells that delete or overwrite data now carry a guard that stops them; their original code is commented out.
- **Future work:** test the prototype on places with known real change; check the pass-count effect on the validation places and flag thin months automatically; re-fetch the 11 WorldCover codes to show what the merge hides; add a case study outside Europe against an independent reference; train several seeds per configuration; consider a temporal input (previous, current and next month) to reduce month-to-month flicker.

Appendix A. Reproducibility

- **Seeds:** sampling 2026, training 42, bootstrap 0, sampler replay 43.
- **Notebooks:** Deep_Learning_final_submission.ipynb is the recorded run of the final project with its saved outputs; Earlier_Attempt/LULC_project_earlier_attempt_v3.ipynb records the first attempt. Both are records and must not be re-run: the cells that would delete or overwrite data are disabled with a guard and their code is commented out. The code of every script used (place selection, download, cache, training, evaluation, seam and false-change tests) is contained in the notebook cells.
- **Dataset:** the folder native_v4 holds 372 places (the first 300 and the 72 extra places) as s1/<place>/<year>-<month>.npz, labels/<place>.npz and meta/<place>.json, with scenes_usable_v4.csv and stats_v4.json. The selection lists are scenes_v4.csv, scenes_v4_extra.csv and pool_v4.csv; scenes_usable_v4b_370_places.csv lists all 370 usable places with their split. The folder is about 80 GB, so download single places.
- **Weights:** best_v4b_fpn_resnet50_imagenet_ctx.pt (about 100 MB) with stats_v4.json, in prototype_v4/weights and in the folder model_weights of the dataset link. The weights of the first attempt are in the second link.
- **Evidence folder:** the saved training histories, configurations, validation and test results and figures that the final notebook reads or produces are in the folder evidence of the submission.

Appendix B. Sources

Sentinel-1 GRD (Earth Engine collection COPERNICUS/S1_GRD; Copernicus Sentinel data, ESA). ESA WorldCover v200 (ESA/WorldCover/v200), the training labels. CORINE Land Cover 2018 (COPERNICUS/CORINE/V20/100m/2018), the case-study reference. Dynamic World (GOOGLE/DYNAMICWORLD/V1), used only for the label-agreement check. RESOLVE Ecoregions 2017 and MODIS land cover, used for sampling. ImageNet weights: torchvision ResNet-50, IMAGENET1K_V1. Software: PyTorch, segmentation_models_pytorch, Streamlit, the Earth Engine Python API, NumPy, pandas and Matplotlib. Map outline in the prototype banner: Natural Earth (public domain).